

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2022

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Tuesday 17th May 2022: 10:00 – 12:30

Answer **ALL** questions

Question 1 carries 35 marks

Question 2 carries 22 marks

Question 3 carries 23 marks

Question 4 carries 20 marks

**THIS EXAMINATION PAPER HAS SIX PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1 – Fluid Mechanics**[35 marks]**

Figure 1 depicts a liquid film being entrained by a plate that moves upwards with a steady velocity U at angle θ with the horizontal. We consider Cartesian coordinates in two dimensions (x, y) where x is aligned with the surface of the plate. Gravity acts vertically downwards. The upward movement of the plate and the opposing action of gravity generate a steady and fully developed flow in the liquid film with velocity $u = u(y)$. The thickness of the liquid film is constant at a value h , and its density ρ and viscosity μ are also constant. The pressure in the surrounding air has a constant value p_0 and you may assume that air has negligible viscosity.

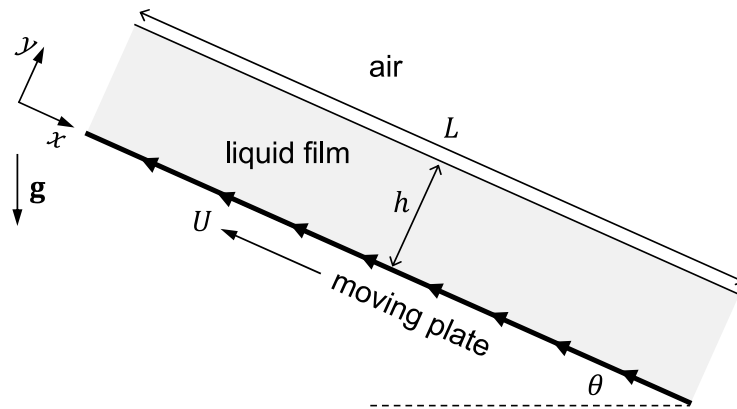


Figure 1. Liquid film entrained by a moving plate.

- (a) Write and simplify the momentum equations in the x and y directions. Clearly state the relevant assumption for each term that can be dropped.

[6 marks]

- (b) Make the simplified momentum equation in the x direction non-dimensional using appropriate scalings. Identify the Reynolds and Froude numbers.

[10 marks]

- (c) Solve for the velocity component, $u = u(y)$, by using the dimensional simplified momentum equation in the x direction obtained in (a). You may assume that the pressure depends only on the y coordinate, i.e., $p = p(y)$.

[10 marks]

- (d) Sketch the velocity and shear stress vector fields within the liquid film.

[9 marks]**END OF QUESTION 1**

Question 2**[22 marks]**

Consider pool boiling of water at 1.0815 bar with a heating platinum wire of diameter 1.5 cm and length 20 cm. The wire is heated uniformly along its length. The water liquid is initially 15 K below its saturation temperature, and the minimum degree of excess temperature above the saturation temperature required for the onset of nucleate boiling is 4 K. The thermophysical properties of saturated water are provided in Table 1 (on page 6).

- (a) Explain which heat transfer regime is present until the onset of nucleate boiling and how you would calculate its heat transfer coefficient (α).

[4 marks]

- (b) At the onset of nucleate boiling, the heat transfer coefficient (α) is $9 \times 10^3 \text{ W m}^{-2} \text{ K}^{-1}$. Estimate the corresponding heat flux. Explain any assumption.

[4 marks]

- (c) List and discuss factors related to the heater surface that may influence the performance of nucleate boiling.

[4 marks]

- (d) Calculate the heat flux and total heat transfer rate from the heating wire to the water pool when the wire surface temperature is at 385 K. Nucleate boiling occurs. The constant depending on the surface-liquid combination (C_{sf}) is 0.0130, and the constant s is 1.0 for water. Explain any assumption.

[5 marks]

- (e) Calculate the heat flux at the Leidenfrost point. Explain any assumption. Reflect on this value in relation to the previous heat flux calculations and results.

[5 marks]**END OF QUESTION 2**

Question 3**[23 marks]**

A superheated steam condenses to liquid water in downward flow as it flows inside a vertical tube of 20 mm diameter. At a certain point along the tube, the steam has a velocity of 10 m s^{-1} , a temperature of 375 K which is 10 K above the saturation temperature, and the condensation rate of 0.004 kg s^{-1} per unit tube length. The overall heat transfer coefficient between the interface of the condensate film and the cooling water on the outside of the tube is $2250 \text{ W m}^{-2} \text{ K}^{-1}$. The thermophysical properties of saturated water are provided in Table 1 (on page 6). You can assume the properties of the superheated vapour are those at the saturation temperature.

- (a) Calculate the total heat flux from the condensing steam to the condensate film. Explain any assumptions.

[15 marks]

- (b) Calculate the temperature of the cooling water.

[5 marks]

- (c) Consider air is mixed with the vapour in a small amount, so the molar fraction of water vapour is slightly below 1. Describe how a process engineer should update the cooling water temperature to keep the same condensation rate.

[3 marks]**END OF QUESTION 3**

Question 4**[20 marks]**

- (a) An air stream is at 20°C dry-bulb temperature, 90% relative humidity and 1 atm. Using the provided psychrometric chart, estimate the following amounts: the humidity, wet-bulb temperature and specific enthalpy of the air stream.

[5 marks]

- (b) 10 kg s⁻¹ of the air stream described in (a) is now mixed with 30 kg s⁻¹ of warm air with a relative humidity of 60% and a dry-bulb temperature of 33°C. What is the resultant dry-bulb temperature?

[5 marks]

- (c) The mixed air stream described in (b) is now dried by using desiccant silica gel to reduce its relative humidity to a value of 20%. Using the provided psychrometric chart, estimate the following amounts:

- (i) Dry-bulb temperature and specific volume of the dried stream. Explain any key assumptions.

[5 marks]

- (ii) The rate at which the desiccant dries the stream.

[5 marks]**END OF QUESTION 4**

Table for Question 2-3:

Table 1. Thermophysical properties of saturated water

Temperature, T (K)	Pressure, p (bar) ^a	Specific Volume (m ³ /kg)		Heat of Vapor- ization, h_{fg} (kJ/kg)	Specific Heat (kJ/kg · K)		Viscosity (N · s/m ²)		Thermal Conductivity (W/m · K)		Prandtl Number		Surface Tension, $\sigma_f \cdot 10^3$ (N/m)
		$v_f \cdot 10^3$	v_g		$c_{p,f}$	$c_{p,g}$	$\mu_f \cdot 10^4$	$\mu_g \cdot 10^4$	$k_f \cdot 10^3$	$k_g \cdot 10^3$	Pr_f	Pr_g	
273.15	0.00611	1.000	206.3	2502	4.217	1.854	1750	8.02	569	18.2	12.99	0.815	75.5
275	0.00697	1.000	181.7	2497	4.211	1.855	1652	8.09	574	18.3	12.22	0.817	75.3
280	0.00990	1.000	130.4	2485	4.198	1.858	1422	8.29	582	18.6	10.26	0.825	74.8
285	0.01387	1.000	99.4	2473	4.189	1.861	1225	8.49	590	18.9	8.81	0.833	74.3
290	0.01917	1.001	69.7	2461	4.184	1.864	1080	8.69	598	19.3	7.56	0.841	73.7
295	0.02617	1.002	51.94	2449	4.181	1.868	959	8.89	606	19.5	6.62	0.849	72.7
300	0.03531	1.003	39.13	2438	4.179	1.872	855	9.09	613	19.6	5.83	0.857	71.7
305	0.04712	1.005	29.74	2426	4.178	1.877	769	9.29	620	20.1	5.20	0.865	70.9
310	0.06221	1.007	22.93	2414	4.178	1.882	695	9.49	628	20.4	4.62	0.873	70.0
315	0.08132	1.009	17.82	2402	4.179	1.888	631	9.69	634	20.7	4.16	0.883	69.2
320	0.1053	1.011	13.98	2390	4.180	1.895	577	9.89	640	21.0	3.77	0.894	68.3
325	0.1351	1.013	11.06	2378	4.182	1.903	528	10.09	645	21.3	3.42	0.901	67.5
330	0.1719	1.016	8.82	2366	4.184	1.911	489	10.29	650	21.7	3.15	0.908	66.6
335	0.2167	1.018	7.09	2354	4.186	1.920	453	10.49	656	22.0	2.88	0.916	65.8
340	0.2713	1.021	5.74	2342	4.188	1.930	420	10.69	660	22.3	2.66	0.925	64.9
345	0.3372	1.024	4.683	2329	4.191	1.941	389	10.89	664	22.6	2.45	0.933	64.1
350	0.4163	1.027	3.846	2317	4.195	1.954	365	11.09	668	23.0	2.29	0.942	63.2
355	0.5100	1.030	3.180	2304	4.199	1.968	343	11.29	671	23.3	2.14	0.951	62.3
360	0.6209	1.034	2.645	2291	4.203	1.983	324	11.49	674	23.7	2.02	0.960	61.4
365	0.7514	1.038	2.212	2278	4.209	1.999	306	11.69	677	24.1	1.91	0.969	60.5
370	0.9040	1.041	1.861	2265	4.214	2.017	289	11.89	679	24.5	1.80	0.978	59.5
373.15	1.0133	1.044	1.679	2257	4.217	2.029	279	12.02	680	24.8	1.76	0.984	58.9
375	1.0815	1.045	1.574	2252	4.220	2.036	274	12.09	681	24.9	1.70	0.987	58.6
380	1.2869	1.049	1.337	2239	4.226	2.057	260	12.29	683	25.4	1.61	0.999	57.6
385	1.5233	1.053	1.142	2225	4.232	2.080	248	12.49	685	25.8	1.53	1.004	56.6

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2023

PART II

and

M.Sc. in ADVANCED CHEMICAL ENGINEERING

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Tuesday 30th May 2023: 09.00 – 11.30

Answer **ALL** questions

Question 1 carries 35 marks

Question 2 carries 22 marks

Question 3 carries 20 marks

Question 4 carries 23 marks

**THIS EXAMINATION PAPER HAS SEVEN PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1 – Fluid Mechanics**[35 marks]**

A solid fibre of radius R_1 is pulled at constant speed U_0 through a cylinder of radius R_2 filled with a liquid of density ρ and viscosity μ , as depicted in Figure 1. The movement of the fibre induces flow in the liquid, which is counteracted by gravity. The resulting flow is assumed to be steady and unidirectional (in cylindrical coordinates, the only non-zero component of the velocity vector is u_z). The pressure gradient in the liquid is assumed to be negligible. Gravity acts vertically downwards.

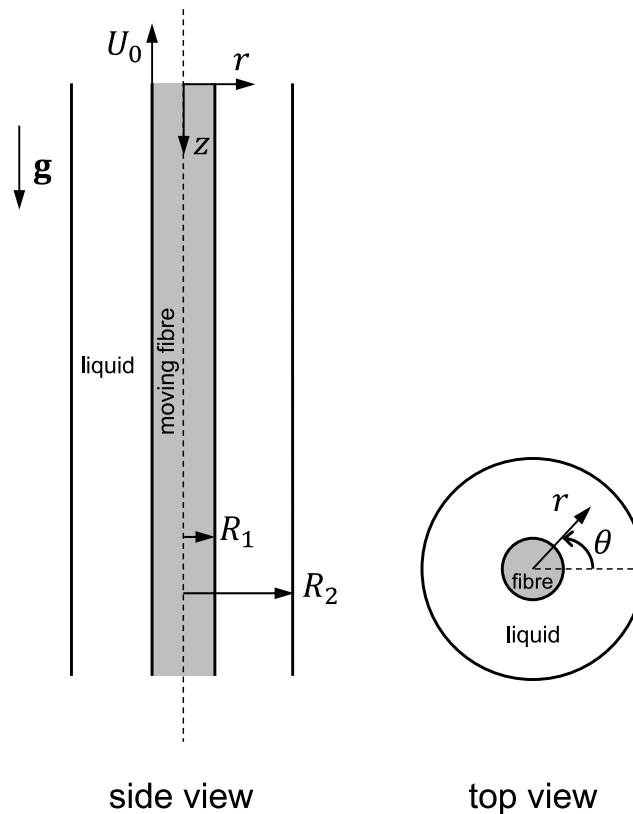


Figure 1 – Liquid flow in a cylinder entrained by a moving fibre.

- (a) How do we refer to this type of flow?

[2 marks]

- (b) Write and simplify the momentum equation in the z direction. Clearly state the relevant assumption for each term that can be dropped.

[8 marks]

- (c) Write the stress tensor matrix for the flow depicted in Figure 1.

[3 marks]

- (d) Make the simplified momentum equation in the z direction non-dimensional using appropriate scalings and identify the relevant dimensionless groups.

[10 marks]

- (e) Solve for the dimensionless velocity component, $\tilde{u}_z(\tilde{r})$, where $\tilde{r}$ is a dimensionless radial coordinate.

[10 marks]

- (f) What is one of the advantages of obtaining the solution of the velocity vector field in dimensionless form?

[2 marks]

END OF QUESTION 1

Question 2**[22 marks]**

- (a) Consider the film boiling regime of water in a pool boiling curve.
- (i) Explain why the total heat transfer coefficient is low near the minimum heat flux point and increases dramatically when the surface temperature gets higher.

[5 marks]

- (ii) Using the physical properties of water at 1.5233 bar in Table 1, calculate the minimum heat flux on a flat horizontal surface.

[5 marks]

- (b) Consider the design of a vertical thermosyphon reboiler. What is the ideal two-phase flow regime at the top of the reboiler, and why?

[5 marks]

- (c) Forced convection correlations usually take the form $Nu = f(Re, Pr)$, where the Nusselt and Reynolds numbers involve a characteristic length (L) and an average velocity (U). Explain Rohsenow's theory that shows that the Nusselt and Reynolds numbers vary as:

$$Nu = \frac{\alpha}{k_l} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \quad Re = \frac{q}{\mu_l h_{lv}} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}}$$

[7 marks]**END OF QUESTION 2**

Question 3**[20 marks]**

In Nusselt analysis of gravity-controlled laminar film condensation on a vertical flat plate, the velocity profile in the liquid film can be expressed as follows:

$$u(y) = \frac{(\rho_l - \rho_v)g\delta^2}{\mu_l} \left[\frac{y}{\delta} - \frac{1}{2} \left(\frac{y}{\delta} \right)^2 \right]$$

- (a) Using the previous velocity profile equation, derive an expression for the mass flow rate per unit width of film, Γ , in terms of local thickness, δ .

[5 marks]

- (b) Considering the energy balance in a differential element of a condensing film, dx , derive an expression for the film thickness, δ , in terms of the vertical distance from the top of the plate, x .

[10 marks]

- (c) Show that the average heat transfer coefficient over the entire plate can be expressed as:

$$\bar{\alpha} = 0.943 \left[\frac{\rho_l(\rho_l - \rho_v)gk_l^3 h_{lv}}{\mu_l(T_{sat} - T_w)L} \right]^{1/4}$$

[5 marks]**END OF QUESTION 3**

Question 4**[23 marks]**

- (a) An air stream with a humidity of $0.0045 \text{ kg (kg dry air)}^{-1}$ has a dry-bulb temperature of 21°C . Use the provided psychrometric chart to determine the relative humidity, the wet-bulb temperature, and the specific enthalpy of the air stream.

[5 marks]

- (b) The above air stream enters a steam-spray humidifier at an equivalent dry air flow rate of 10 kg s^{-1} . Saturated steam at 1 atm having a specific enthalpy of 2676 kJ kg^{-1} and a flow rate of 0.020 kg s^{-1} is injected into the air stream so that the air leaves with a higher humidity. You can assume that there is no heat transfer with the surroundings, and the pressure is constant throughout at 1 atm.

- (i) What is the specific enthalpy of the final stream?

[5 marks]

- (ii) What the final relative humidity?

[5 marks]

- (iii) What is the maximum saturated steam flow rate that can be injected before condensation occurs?

[5 marks]

- (c) How would you increase the dry bulb temperature of the resulting air stream keeping a constant specific enthalpy?

[3 marks]**END OF QUESTION 4**

Table for Question 2:

Table 1. Thermophysical properties of saturated water

Temperature, T (K)	Pressure, p (bar) ^b	Specific Volume (m ³ /kg)		Heat of Vapor- ization, h_{fg} (kJ/kg)	Specific Heat (kJ/kg · K)		Viscosity (N · s/m ²)		Thermal Conductivity (W/m · K)		Prandtl Number		Surface Tension, $\sigma_f \cdot 10^3$ (N/m)
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IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2024

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Monday, 13 May 2024: 09.00 – 11.30

Answer **ALL** questions

Question 1 carries 35 marks

Question 2 carries 23 marks

Question 3 carries 26 marks

Question 4 carries 16 marks

**THIS EXAMINATION PAPER HAS 9 PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1 – Fluid Mechanics**[35 marks]**

Fluid of density ρ and viscosity μ flows between two parallel plates under the influence of a constant pressure gradient dp/dx , as shown in Figure 1. Midway between the plates there is a slab b of stationary porous material with permeability k . The flow is assumed to be steady and two-dimensional, yielding the velocity profile, $u(y)$, depicted in the diagram. Gravity can be neglected. The fluid velocity within the porous slab is given by the following expression:

$$u_b = -\frac{k}{\mu} \frac{dp}{dx}$$

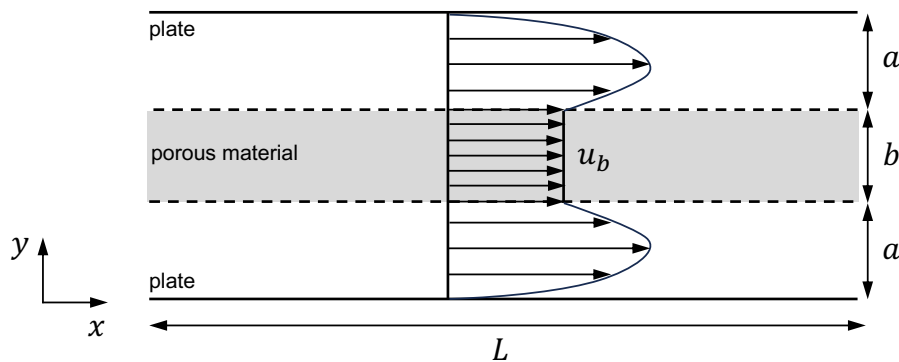


Figure 1. Liquid flow between two parallel plates and a porous material.

- (a) You shall apply the lubrication approximation to obtain the velocity profile in gap a . Describe the condition for which this approximation is valid.

[2 marks]

- (b) Identify the characteristic length scales and velocity scales of the problem, and define the associated dimensionless variables.

[6 marks]

- (c) Apply the lubrication approximation to simplify the momentum equation in the x direction. Clearly state the relevant assumption for each term that can be dropped and the chosen pressure scale. You shall assume that $Re = \rho u_b L / \mu \approx 1$.

[12 marks]**QUESTION 1 CONTINUES**

- (d) Show that the velocity profile for $0 \leq y \leq a$ is given by:

$$u = \frac{1}{\mu} \left(-\frac{dp}{dx} \right) \left[\frac{1}{2} (ay - y^2) + \frac{yk}{a} \right]$$

[10 marks]

- (e) Compute the shear stress of the fluid in contact with the lower boundary of the porous slab.

[5 marks]

END OF QUESTION 1

Question 2 – Boiling

[23 marks]

(a) Consider the data extracted from Berenson's study¹ on the influence of surface conditions on pool boiling in Figure 2.

- i. Identify which data (circles or squares) pertain to the roughest surface conditions and justify your answer.

[4 marks]

- ii. Briefly explain the different pool boiling regimes and how these are impacted by surface conditions.

[6 marks]

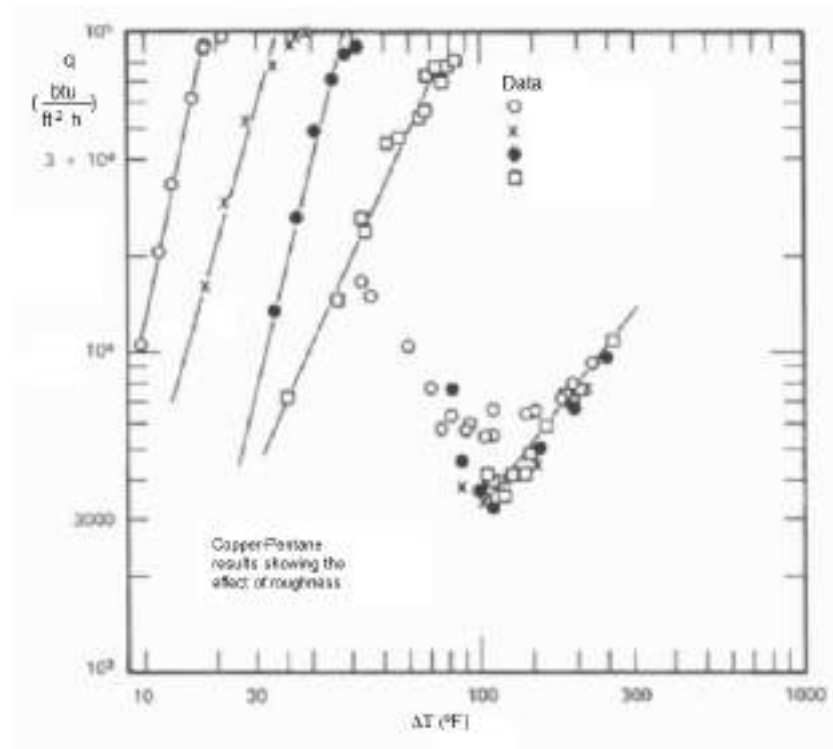


Figure 2: Data extracted from Berenson's study¹ of the influence of surface conditions on the boiling curve.

QUESTION 2 CONTINUES

¹ Berenson P. J. 1962 Experiments on pool-boiling heat transfer. Int J Mass Transfer, 5, 985-999

- (b) Consider a kettle reboiler connected to a distillation column, where a mixture needs to be partially vaporised. The reboiler operates at a pressure of 0.9040 bar and the working fluid is mainly water. The heating fluid is steam. The design specifications require that the reboiler must transfer a heat load, Q , of 500 kW without reaching the critical heat flux condition.
- i. Using the physical properties of water in Table 1, calculate the minimum required heating surface area of the reboiler to ensure operation below the critical heat flux condition.
- [5 marks]**
- ii. Estimate the maximum steam operating pressure to avoid the critical heat flux condition. You shall assume the following values of the constants in the Rohsenow's equation: $C_{sf} = 0.0130$ and $s=1.0$ for water.
- [5 marks]**
- iii. Estimate the mass flow rate of vapour generated, assuming that the heat load is fully utilised for vaporisation.
- [3 marks]**

Note Table 1 of thermophysical properties of saturated water in page 9.

END OF QUESTION 2

Question 3 – Condensation**[26 marks]**

- (a) Consider laminar film condensation at high vapour velocities, where the interfacial shear becomes significant compared to the gravitational force on the liquid phase. Write the momentum equation in the x direction for the liquid phase (condensate film) and demonstrate the following:

- i. The mean velocity in the condensate film is given by:

$$\bar{u} = \frac{\tau_i}{2\mu_l} \delta$$

where δ is the film thickness, τ_i is the shear stress at the liquid-vapor interface and μ_l is the liquid viscosity. Explain any assumptions.

[6 marks]

- ii. The local heat transfer coefficient is given by:

$$\alpha = \sqrt{\frac{k_l^2 \tau_i \rho_l}{2\Gamma \mu_l}}$$

where k_l is the liquid thermal conductivity, Γ is the condensate mass flow rate per unit width, and ρ_l is the liquid density.

[6 marks]

- (b) Now, consider a scenario where a mixture of air and water vapour is flowing through a tube with a diameter of 5 cm. The mixture, moving at a speed of 30 m s^{-1} is undergoing condensation. The weight percentage of water vapour in the gas mixture is 20%, the gas temperature is 180°C , and the absolute pressure is 1.5 bar.

- i. Illustrate a diagram depicting how the partial pressures of air and water vapour vary along the distance from the interface between the gas and liquid phases.

[4 marks]

- ii. Determine the partial pressure of water vapour at the gas-liquid interface, given that the rate of water condensation is 0.002 kg s^{-1} per meter length of tube. Assume that mass transfer in the tube can be described by the following specific correlation:

$$Sh = 0.023 \times Re^{0.8} \times Sc^{1/3}$$

QUESTION 3 CONTINUES

The physical properties of water in Table 1 and the following physical properties for the gas mixture may be used:

Data for Question 3d:

Molar mass of air	M_{air}	28.96	g mol^{-1}
Molar mass of water	M_{water}	18.02	g mol^{-1}
Gas density	ρ_v	0.744	kg m^{-3}
Gas viscosity	μ_v	2.57×10^{-5}	$\text{kg m}^{-1} \text{s}^{-1}$
Binary diffusion coefficient	D_{12}	5.12×10^{-5}	$\text{m}^2 \text{s}^{-1}$
Gas constant	R	8.314	$\text{J mol}^{-1} \text{K}^{-1}$

[10 marks]

END OF QUESTION 3

Question 4 – Air-Water Systems**[16 marks]**

- (a) An air stream has a relative humidity of 50% and a dry-bulb temperature of 25°C. Use the provided psychrometric chart to determine the humidity, wet-bulb temperature, and dew-point temperature of the air stream.

[3 marks]

- (b) 10 kg h⁻¹ of saturated air at 16°C is heated up in a heating coil until the relative humidity decreases to 30%. This drier and warmer air is then used in a drying process to extract moisture from a wet solid. The air leaves the drying process with a relative humidity of 60%.

- i. What is the temperature of the air after heating?

[3 marks]

- ii. Assuming negligible heat loss to the surroundings, estimate the heating power required.

[5 marks]

- iii. Calculate the rate of water extraction in the drying process.

[5 marks]**END OF QUESTION 4**

Table for Question 2 and 3d:

Table 1. Thermophysical properties of saturated water

TABLE A.6 Thermophysical Properties of Saturated Water^a

Temperature, T (K)	Pressure, p (bar) ^b	Specific Volume (m ³ /kg)		Heat of Vaporization, h_{fg} (kJ/kg)	Specific Heat (kJ/kg · K)		Viscosity (N · s/m ²)		Thermal Conductivity (W/m · K)		Prandtl Number		Surface Tension, σ_s · 10 ³ (N/m)	Expansion Coefficient, β · 10 ⁶ (K ⁻¹)	Temperature, T (K)
		v_f · 10 ³	v_g		$c_{p,f}$	$c_{p,g}$	μ_f · 10 ⁶	μ_g · 10 ⁶	k_f · 10 ³	k_g · 10 ³	Pr_f	Pr_g			
273.15	0.00611	1.000	206.3	2502	4.217	1.854	1750	8.02	569	18.2	12.99	0.815	75.5	-68.05	273.15
275	0.00687	1.000	181.7	2497	4.211	1.855	1652	8.09	574	18.3	12.22	0.817	75.3	-32.34	275
280	0.00990	1.000	130.4	2485	4.198	1.858	1422	8.29	582	18.6	10.26	0.825	74.8	46.04	280
285	0.01387	1.000	99.4	2473	4.189	1.861	1225	8.49	590	18.9	8.81	0.833	74.3	114.1	285
290	0.01917	1.001	69.7	2461	4.184	1.864	1080	8.69	598	19.3	7.56	0.841	73.7	174.0	290
295	0.02617	1.002	51.94	2449	4.181	1.868	959	8.89	606	19.5	6.62	0.849	72.7	227.5	295
300	0.03571	1.003	39.13	2438	4.179	1.872	855	9.09	613	19.6	5.83	0.857	71.7	276.1	300
305	0.04712	1.005	29.74	2426	4.178	1.877	768	9.29	620	20.1	5.20	0.865	70.9	320.6	305
310	0.06221	1.007	22.93	2414	4.178	1.882	695	9.49	628	20.4	4.62	0.873	70.0	361.9	310
315	0.08132	1.009	17.82	2402	4.179	1.888	631	9.69	634	20.7	4.14	0.881	69.2	400.4	315
320	0.1053	1.011	13.98	2390	4.180	1.895	577	9.89	640	21.0	3.77	0.894	68.3	436.7	320
325	0.1351	1.013	11.06	2378	4.182	1.903	528	10.09	645	21.3	3.42	0.901	67.5	471.3	325
330	0.1719	1.016	8.82	2366	4.184	1.911	489	10.29	650	21.7	3.15	0.908	66.6	504.0	330
335	0.2167	1.018	7.09	2354	4.186	1.920	453	10.49	656	22.0	2.88	0.916	65.8	535.5	335
340	0.2713	1.021	5.74	2342	4.188	1.930	420	10.69	660	22.3	2.66	0.925	64.9	566.0	340
345	0.3372	1.024	4.683	2329	4.191	1.941	389	10.89	664	22.6	2.45	0.933	64.1	595.4	345
350	0.4163	1.027	3.846	2317	4.195	1.954	365	11.09	668	23.0	2.29	0.942	63.2	624.2	350
355	0.5100	1.030	3.180	2304	4.199	1.968	343	11.29	671	23.3	2.14	0.951	62.3	652.3	355
360	0.6208	1.034	2.645	2291	4.203	1.983	324	11.49	674	23.7	2.02	0.960	61.4	679.9	360
365	0.7514	1.038	2.232	2278	4.209	1.999	306	11.69	677	24.1	1.91	0.969	60.5	707.1	365
370	0.9040	1.041	1.861	2266	4.214	2.017	289	11.89	679	24.5	1.80	0.978	59.5	728.7	370
375.15	1.0133	1.044	1.679	2257	4.217	2.029	275	12.02	680	24.8	1.76	0.984	58.9	750.1	375.15
375	1.0815	1.045	1.574	2252	4.220	2.036	274	12.09	681	24.9	1.70	0.987	58.6	761	375
380	1.2869	1.049	1.337	2239	4.226	2.057	260	12.29	683	25.4	1.61	0.999	57.6	788	380
385	1.5233	1.053	1.142	2225	4.232	2.080	248	12.49	685	25.8	1.53	1.004	56.6	814	385
390	1.794	1.058	0.980	2212	4.239	2.104	237	12.69	686	26.3	1.47	1.013	55.6	841	390
400	2.455	1.067	0.731	2183	4.256	2.158	217	13.05	688	27.2	1.34	1.033	53.6	896	400
410	3.302	1.077	0.553	2153	4.278	2.221	200	13.42	688	28.2	1.24	1.054	51.5	952	410
420	4.370	1.088	0.425	2123	4.302	2.291	185	13.79	688	29.8	1.16	1.075	49.4	1010	420
430	5.699	1.099	0.331	2091	4.331	2.369	173	14.14	685	30.4	1.09	1.10	47.2		430

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2025

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Monday 12th May 2025: 09.00 – 11.30

Answer **ALL** four questions

Question 1 carries 35 marks

Question 2 carries 24 marks

Question 3 carries 24 marks

Question 4 carries 17 marks

**THIS EXAMINATION PAPER HAS EIGHT PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1**[35 marks]**

Consider steady unidirectional, incompressible flow of two immiscible liquids between two parallel stationary plates, which are separated by a distance, H , as shown in Figure 1. The two liquids, A and B, have densities ρ_A and ρ_B , and viscosities μ_A and μ_B , respectively. The interface between the two liquids is located at $y = d$. The flow is fully developed, and gravity may be considered unimportant. To solve this problem, you shall use two-dimensional Cartesian coordinates (x, y) with corresponding velocity components (u, v) and pressure p .

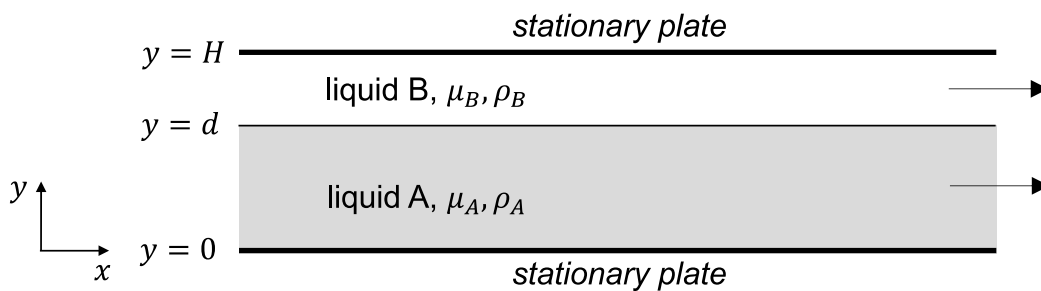


Figure 1 – Geometry for the flow of two immiscible liquids between two parallel plates.

- (a) Verify that the pressure p is only a function of the horizontal distance, x .

[6 marks]

- (b) Considering layer A ($0 \leq y \leq d$), simplify the x –component momentum equation, clearly stating the relevant assumption for each term that can be dropped.

[5 marks]

- (c) Integrate the differential equation obtained in (b) to yield the velocity component u_A in terms of y and any or all of $d, dp/dx, \mu_A, \rho_A$, and two arbitrary integration constants, say, c_{1A} and c_{2A} . You shall assume that a similar relationship holds for u_B .

[4 marks]**QUESTION 1 CONTINUES**

- (d) State all the boundary conditions and derive expressions for the four integration constants, to obtain expressions for the velocity profiles in the two liquids, $u_A(y)$ and $u_B(y)$.

Hint: in recasting the equations, you may find helpful to refer to the analytical solution for a single-fluid, i.e.

$$u = \frac{1}{2\mu} \left(\frac{dp}{dx} \right) H^2 \left[\left(\frac{y}{H} \right)^2 - \left(\frac{y}{H} \right) \right]$$
$$\tau_{yx} = \frac{dp}{dx} \left(y - \frac{H}{2} \right)$$

[10 marks]

- (e) Obtain the expression for the shear-stress profile between the upper and lower plates.

[5 marks]

- (f) Make a qualitative sketch of the velocity and stress profile between the two plates for the situation $d = H/2$ and $\mu_B/\mu_A = 2$. Clearly indicate the following features of the profiles: (i) location of the maximum in the velocity; (ii) location of zero-stress, and (iii) which plate experiences the largest shear stress.

[5 marks]

END OF QUESTION 1

Question 2 – Boiling**[24 marks]**

In a kettle-type reboiler, water is heated by a bundle of 20 stainless-steel tubes of external diameter 4 cm and 5 m length. The water in the reboiler is saturated at 1 atm, and the vapour formed at the pool is fed to a distillation column.

- (a) Consider pool boiling of the water in the reboiler:
- (i) Draw the kettle-type reboiler and explain why pool boiling is typically considered in this type of reboilers.
 - (ii) Explain the pool boiling curve and identify the different boiling regimes. Describe the heat transfer mechanisms in each regime.
 - (iii) Why is the burnout point (critical heat flux) typically avoided in the design of boilers?

[9 marks]

- (b) The surface temperature of the tubes has been unintentionally overheated to 380 °C. The emissivity of the steel tube surface is 0.8. Using the physical properties of water vapour in Table 1 and saturated water in Table 2 located at the end of this paper, calculate the following:
- (i) Average heat transfer coefficient.
 - (ii) Total heat transfer rate from the tubes to the pool. Using Table 3, reflect on how this value would change if stainless steel were replaced with nickel.

[15 marks]**END OF QUESTION 2**

Question 3 – Condensation**[24 marks]**

Consider laminar film condensation of a stagnant water vapour at 1 atm on a vertical plate of 30 cm that is maintained at 95 °C.

- (a) Write the momentum equation in the direction of the moving liquid phase (condensate film) and eliminate any negligible terms. Explain your assumptions

[8 marks]

- (b) Using the previous derived momentum equation, derive an equation for the velocity of the condensate as a function of the condensate thickness.

[4 marks]

- (c) Using the thermophysical properties in Table 2, calculate the average velocity of the condensate at the bottom of the plate. You may use the following expression for the

condensate thickness $\delta(x) = \left[\frac{4k_l \mu_l (T_{sat} - T_w)x}{\rho_l (\rho_l - \rho_v) g h_{lv}} \right]^{1/4}$. Discuss what would happen if

the plate was very long.

[8 marks]

- (d) Calculate the heat transfer flux at the bottom of the plate

[4 marks]**END OF QUESTION 3**

Question 4 – Air-water mixtures

[17 marks]

- (a) A stream of air at 1 atm, 40°C and 30% relative humidity needs to be brought to the "comfort" conditions of 22°C and 60% relative humidity by mixing with a second air stream. The mass flow rate of the first air stream before mixture is 50 kg h⁻¹. Using the provided psychrometric chart, estimate
- (i) The humidity, wet-bulb temperature, and dew-point temperature of the first air stream. [3 marks]
- (ii) The minimum mass flow rate of the second air stream. [5 marks]
- (iii) Could evaporative cooling be used instead of air mixing to achieve the desired conditions? [4 marks]
- (b) Inside a pipe, water and air flow in opposite directions, creating an annular flow pattern. The air is at 30°C with 30% relative humidity, while the water is at 20°C, as shown in Figure 2. Using the provided psychrometric chart, what is the minimum temperature at which the air can exit the pipe?

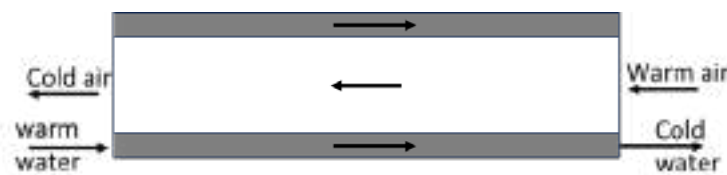


Figure 2 – Schematic of a pipe with warm water and cold air in counterflow.

[5 marks]

END OF QUESTION 4

Tables for Question 2:

Table 1. Thermophysical properties of water vapour at 1 atm

T (K)	ρ (kg/m ³)	c_p (J/kg·K)	μ (kg/m·s)	ν (m ² /s)	k (W/m·K)	α (m ² /s)	Pr
373.15	0.5976	2080	12.28×10^{-6}	20.55×10^{-6}	0.02509	2.019×10^{-5}	1.018
393.15	0.5652	2021	13.04	23.07	0.02650	2.320	0.994
413.15	0.5365	1994	13.81	25.74	0.02805	2.622	0.982
433.15	0.5108	1980	14.59	28.56	0.02970	2.937	0.973
453.15	0.4875	1976	15.38	31.55	0.03145	3.265	0.966
473.15	0.4665	1976	16.18	34.68	0.03328	3.610	0.961
493.15	0.4472	1980	17.00	38.01	0.03519	3.974	0.956
513.15	0.4295	1986	17.81	41.47	0.03716	4.357	0.952
533.15	0.4131	1994	18.63	45.10	0.03919	4.758	0.948

Table 2. Thermophysical properties of saturated water

Temperature, T (K)	Pressure, p (bar) ^a	Specific Volume (m ³ /kg)		Heat of Vaporization, h_{fg} (kJ/kg)	Specific Heat (kJ/kg·K)		Viscosity (N·s/m ²)		Thermal Conductivity (W/m·K)		Prandtl Number		Surface Tension, $\sigma_f \cdot 10^3$ (N/m)	Expansion Coefficient, $\beta_f \cdot 10^6$ (K ⁻¹)	Temperature, T (K)
		$v_f \cdot 10^{-3}$	v_g		$c_{p,f}$	$c_{p,g}$	$\mu_f \cdot 10^{-4}$	$\mu_g \cdot 10^{-4}$	$k_f \cdot 10^{-3}$	$k_g \cdot 10^{-3}$	Pr_f	Pr_g			
273.15	0.00611	1.000	206.3	2502	4.217	1.854	1750	8.02	569	18.2	12.99	0.815	75.5	-68.05	273.15
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340	0.2713	1.021	5.74	2342	4.188	1.930	420	10.69	660	22.3	2.66	0.923	64.9	566.0	340
345	0.3372	1.024	4.683	2329	4.191	1.941	389	10.89	664	22.6	2.45	0.931	64.1	595.4	345
350	0.4163	1.027	3.846	2317	4.195	1.954	365	11.09	668	23.0	2.29	0.942	63.2	624.2	350
355	0.5106	1.030	3.180	2304	4.199	1.968	343	11.29	671	23.3	2.14	0.951	62.3	652.3	355
360	0.6209	1.034	2.645	2291	4.203	1.983	324	11.49	674	23.7	2.02	0.960	61.4	679.9	360
365	0.7514	1.038	2.212	2278	4.209	1.999	306	11.69	677	24.1	1.91	0.969	60.5	707.1	365
370	0.9040	1.041	1.861	2265	4.214	2.017	289	11.89	679	24.5	1.80	0.978	59.5	728.7	370
373.15	1.0133	1.044	1.679	2257	4.217	2.029	279	12.02	680	24.8	1.76	0.984	58.9	750.1	373.15
375	1.0815	1.045	1.574	2252	4.220	2.036	274	12.09	681	24.9	1.70	0.987	58.6	761	375
380	1.2869	1.049	1.337	2239	4.226	2.057	260	12.29	683	25.4	1.61	0.999	57.6	788	380
385	1.5233	1.053	1.142	2225	4.232	2.080	248	12.49	685	25.8	1.53	1.004	56.6	814	385
390	1.794	1.058	0.980	2212	4.239	2.104	237	12.69	686	26.3	1.47	1.013	55.6	841	390
400	2.455	1.067	0.731	2180	4.256	2.158	217	13.05	688	27.2	1.34	1.033	53.6	896	400
410	3.302	1.077	0.553	2153	4.278	2.221	200	13.42	688	28.2	1.24	1.054	51.5	952	410
420	4.370	1.088	0.425	2123	4.302	2.291	185	13.79	688	29.8	1.16	1.075	49.4	1010	420
430	5.699	1.099	0.331	2091	4.331	2.369	173	14.14	685	30.4	1.09	1.10	47.2		430

Table 3. Metal surface emissivity

Metals		ϵ
Aluminum	polished	0.039
	oxidized	0.200
Brass	polished	0.030
	oxidized	0.610
Chromium	polished	0.075
Copper	polished	0.023
	oxidized	0.780
Gold	polished	0.020
Iron	polished	0.210
	oxidized	0.740
	tin-plated	0.070
Lead	polished	0.060
	oxidized	0.630
Nickel	polished	0.072
	oxidized	0.350
Steel	polished stainless	0.074
	oxidized	0.790
Titanium		0.100
Tungsten		0.110